19th International Mathematics and Science Olympiad

SHORT ANSWER PROBLEMS

ANSWER SHEET

Name : _____ **Index Number** : _____

Country : _____ **Score** : _____

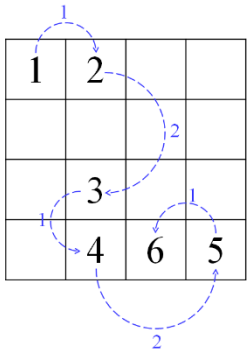
1.	6 grandchildren	**14.**	52 blocks
2.	3 chairs	**15.**	51319
3.	65	**16.**	875 judges
4.	2045254	**17.**	14 cm
5.	$150\sqrt{3}\,\text{m}^2$	**18.**	75
6.	\$450	**19.**	36
7.	23 litres	**20.**	7 cm
8.	3047910	**21.**	22
9.	40°	**22.**	3478 steps
10.	25	**23.**	60
11.	18	**24.**	39
12.	$16\ \text{cm}^2$	**25.**	95
13.	699		

19th IMSO Essay Answers (2022)

1.	1111	**2.**	112	**3.**	45
4.	22	**5.**	30	**6.**	48
7.	1996	**8.**	If the biggest orange that we took in every row is in one column, so, this will make $X = Y$ regarding to the condition, so the answer ***NO***. If the biggest orange that we took in every row is not in one column, the answer is ***YES*** and $X > Y$		
9.	$65° < B + C < 75°$ when $40° < \angle AMH < 60°$.			**10.**	114
11.	28	**12.**	67	**13.**	135

SOLUTIONS OF EXPLORATION PROBLEMS

1. In the diagram below, there is a 4 × 4 grid with a chain of consecutive positive integers starting from 1; i.e. 1 → 2 → 3 → 4 → 5 → 6, where the distances towards the next number are 1, 2, 1, 2, 1 sequentially (1 and 2 alternately).



Construct a chain of consecutive positive integers **as long as possible** in both grids below, starting from 1 at the top left corner and the distances towards the next number being 1, 2, 1, 2, … in the alternating manner.

What is maximum number that can be placed on a:

(i) 4 x 4 grid ? Draw one example.

(ii)5 x 5 grid? Draw one example.

[*Singapore*]

The table format is incorrect:

1			

(a) 4 × 4 table

(2 points)

1				

(b) 5 × 5 table

(4 points)

[Solution]

The longest chain for a 4 × 4 table is a chain with length 16.

The longest chain for a 5 × 5 table is a chain with length 25.

There are many ways to construct the longest chain.

Examples of both tables are shown below.

1	2	16	3
6	5	15	4
12	11	13	10
7	8	14	9

1	2	4	3	5
10	9	7	8	6
20	19	17	18	16
11	12	14	13	15
21	22	24	23	25

[Marking Scheme]

- **(a) 2 marks or 0 for correct answer and correct drawing, no partial score.**
- **(b) 4 marks or 0 for correct answer and correct drawing, no partial score.**

2. Construct five different rectangles of various sizes by selecting each of the numbers from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 exactly once to be the length and width dimensions.

   By using these five rectangles, can these pieces be tiled (without overlapping) to form:

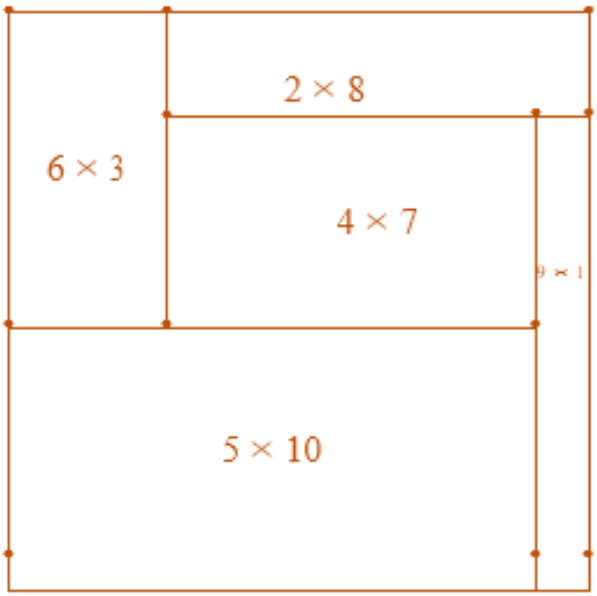
   (I) a 11 x 11 square? If yes, draw one such example. If not, state the reason why it is not possible.

   (II) a 13 x 13 square? If yes, draw one such example. If not, state the reason why it is not possible.

   Note that all rectangles formed can be rotated or reflected. **[*USA*]**

[Solution]

Yes, both 11×11 and 13×13 are tile able using such rectangles. For the 13×13 square are $12, 45, 38, 79, 610$ and $12, 46, 37, 89, 510$, and for the 11×11 square are $16, 47, 58, 39, 210$ and $36, 47, 28, 19, 510$ work.



[Marking Scheme]

- **Give one correct figure for (i) …… 3 marks**
- **Give one correct figure for (ii) …….. 3 marks**
- **Given Yes/no answers only without any figures drawn ….. 0 mark**
- **Incorrect figures drawn …….. 0 mark**

3. In the first move, we can make a straight cut across a square paper, dividing it into two smaller polygons. In each subsequent move, we can divide any one of the resulting polygons into two with a straight cut. No folding while cutting and no two papers can cut together.
   a. What is the minimum number of cuts required to obtain 2 polygons each with exactly 20 sides, along with some other polygons?
   b. What is the minimum number of cuts required to obtain 3 polygons each with exactly 30 sides, along with some other polygons?
   c. What is the minimum number of cuts required to obtain 4 polygons each with exactly 60 sides, along with some other polygons?

[*Taiwan*]

[Solution]

a. First make 1 cut to create 2 rectangles. From each rectangle, cut off triangles at the vertices 16 times to obtain a polygon with exactly 20 sides. The total number of cuts is $1+2\times16=33$

b. First make 2 cuts to create 3 rectangles. From each rectangle, cut off triangles at the vertices 26 times to obtain a polygon with exactly 30 sides. The total number of cuts is $2+3\times26=80$.

c. First make 3 cuts to create 4 rectangles. From each rectangle, cut off triangles at the vertices 56 times to obtain a polygon with exactly 60 sides. The total number of cuts is $3+4\times56=227$.

Answer:

a. 33 cuts

b. 80 cuts

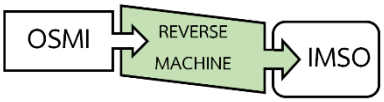
c. 227 cuts

[Marking Scheme]

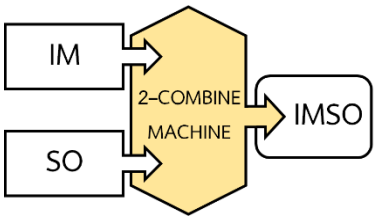
- **Correct Solution for part a ……. 1 mark**
- **Correct Solution for part b ……. 2 marks**
- **Correct Solution for part c ……. 3 marks**

4. We have three kinds of machines namely “REVERSE”, “2-COMBINE” and “3-COMBINE” Machines.

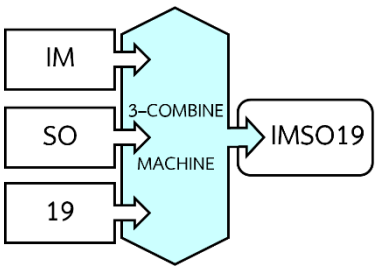
   A “REVERSE MACHINE” will reverse the order of the letters of the input as shown below:



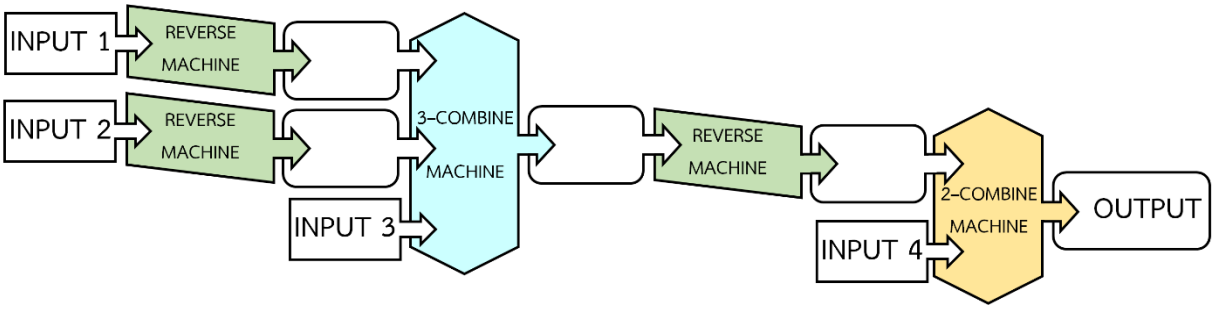
   A “2–COMBINE MACHINE” will connect 2 inputs together (putting the upper input in front of the lower input) as shown below:



   A “3–COMBINE MACHINE” will connect 3 inputs together (putting the upper-most input in front of the middle input and then followed by the lower-most input) as shown below:



   The following system consists of three “REVERSE MACHINES”, one “2–COMBINE MACHINE” and one “3–COMBINE MACHINE”.



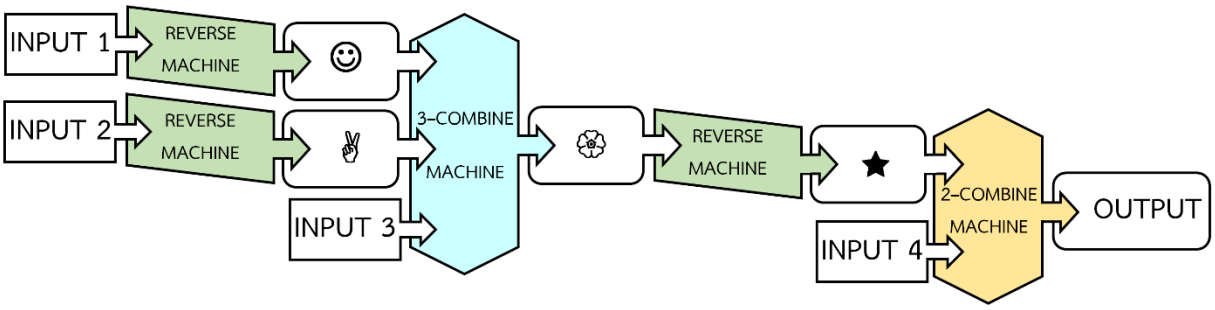
   Renard’s comment: machine system figure first before problems.

   a. Give TWO (2) examples for “INPUT 1”, “INPUT 2”, “INPUT 3” and “INPUT 4” such that the “OUTPUT” will be “ILOVEIMSO19”.

   b. How many different arrangements can we have for “INPUT 1”, “INPUT 2”, “INPUT 3” and “INPUT 4” such that the “OUTPUT” will be “ILOVEIMSO19”? (Note that each input must have at least 1 letter.) **[*Thailand*]**

[Solution]

It will be easy if we work backward. First, we focus on "INPUT 4". Then, we know "★" and "❁".



INPUT 4	9	19	O19	SO19	MSO19	IMSO19	EIMSO19	VEIMSO19
★	ILOVEIMSO1	ILOVEIMSO	ILOVEIMS	ILOVEIM	ILOVEI	ILOVE	ILOV	ILO
❁	1OSMIEVOLI	OSMIEVOLI	SMIEVOLI	MIEVOLI	IEVOLI	EVOLI	VOLI	OLI

By the same idea, we can separate the case of "INPUT 3" and "✌" for each "INPUT 4".

Then "☺", "INPUT 1" and "INPUT 2" will be fixed.

INPUT 4	9	19	O19	SO19	MSO19	IMSO19	EIMSO19	VEIMSO19
★	ILOVEIMSO1	ILOVEIMSO	ILOVEIMS	ILOVEIM	ILOVEI	ILOVE	ILOV	ILO
❁	1OSMIEVOLI	OSMIEVOLI	SMIEVOLI	MIEVOLI	IEVOLI	EVOLI	VOLI	OLI
✌, **INPUT3**	OSMIEVOL, I	SMIEVOL, I	MIEVOL, I	IEVOL, I	EVOL, I	VOL, I	OL, I	L, I
	OSMIEVO, LI	SMIEVO, LI	MIEVO, LI	IEVO, LI	EVO, LI	VO, LI	O, LI	
	OSMIEV, OLI	SMIEV, OLI	MIEV, OLI	IEV, OLI	EV, OLI	V, OLI		
	OSMIE, VOLI	SMIE, VOLI	MIE, VOLI	IE, VOLI	E, VOLI			
	OSMI, EVOLI	SMI, EVOLI	MI, EVOLI	I, EVOLI				
	OSM, IEVOLI	SM, IEVOLI	M, IEVOLI					
	OS, MIEVOLI	S, MIEVOLI						
	O, SMIEVOLI							

If INPUT 4 is "9", there are 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 36 ways to obtain the desired result.

If INPUT 4 is "19", there are 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28 ways to obtain the desired result.

If INPUT 4 is "O19", there are 6 + 5 + 4 + 3 + 2 + 1 = 21 ways to obtain the desired result.

If INPUT 4 is "SO19", there are 5 + 4 + 3 + 2 + 1 = 15 ways to obtain the desired result.

If INPUT 4 is "MSO19", there are 4 + 3 + 2 + 1 = 10 ways to obtain the desired result.

If INPUT 4 is "IMSO19", there are 3 + 2 + 1 = 6 ways to obtain the desired result.

If INPUT 4 is "EIMSO19", there are 2 + 1 = 3 ways to obtain the desired result.

If INPUT 4 is "VEIMSO19", there are 1 way to obtain the desired result.

Hence, there are 36 + 28 + 21 + 15 + 10 + 6 + 3 + 1 = 120 ways to put the "INPUT 1", "INPUT 2", "INPUT3" and "INPUT 4" so that the "OUTPUT" will be "ILOVEIMSO19"

Answer: 120 ways

Answer: 120 ways

[Marking Scheme]

(a) Each correct example ..... 0,5 mark

(b) Correct Complete answer with example ..... 5 marks

- **Correct answer without proper explanation for part b) ..... 1 mark**

5. Given a sequence such that each number only contains odd digits:

1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 31, 33, 35, 37, 39, 51, 53, 55, …

a) Find the 2022nd term.

b) Find the smallest term such that its sum of digits is 2022.

c) Find the sum of all terms that does not exceed 2022.

[*Vietnam*]

[Solution]

a) There are 5 numbers $\overline{a}$ that has an odd digit: 1, 3, 5, 7, 9.

The number of $\overline{ba}$ that satisfies a, b are odd digits is: $5 \times 5 = 25$ numbers.

The number of $\overline{cba}$ that satisfies a, b, c are odd digits is: $5 \times 5 \times 5 = 125$ numbers.

The number of $\overline{dcba}$ that satisfies a, b, c, d are odd digits is:

$5 \times 5 \times 5 \times 5 = 625$ numbers.

The number of $\overline{edcba}$ that satisfies a, b, c, d, e are odd digits is:

$5 \times 5 \times 5 \times 5 \times 5 = 3125$ numbers.

Therefore there are $5 + 25 + 125 + 625 = 780$ numbers having from 1 to 4 odd digits.

We need $2022 - 780 = 1242$ more 5-digit numbers.

The 780th number is 9999

The 780 + 625 = 1405th number is 19999

The 1405 + 625 = 2030th number is 39999

We need to step back 2030 – 2022 = 8 numbers with all odd digits starting from 39999.

These 8 numbers are: 39997, 39995, 39993, 39991, 39979, 39977, 39975, 39973.

Thus the 2022th number is 39973.

b) Let A be the smallest number with n digits such that all of its digits are odd, and the sum of them is 2022.

Since the sum of n odd digits are an even number 2022, n must be even.

Because 224 × 9 = 2016 < 2022, so A must have 226 digits.

Thus A = 15999…99, with 224 consecutive digits 9.

c) The sum of all numbers in the sequence with one digit is: $1 + 3 + 5 + 7 + 9 = 25$.

The sum of all numbers in the sequence with two digits is:

$5 \times (1 + 3 + 5 + 7 + 9) \times 11 = 1375.$

The sum of all numbers in the sequence with three digits is:

$25 \times (1 + 3 + 5 + 7 + 9) \times 111 = 69375.$

Notice that the largest number in the sequence not exceeding 2022 is 1999.

The sum of all numbers in the sequence with four digits is:

$69375 + 125 \times 1000 = 194375.$

Thus the sum of all numbers in the sequence not exceeding 2022 is:

$25 + 1375 + 69375 + 194375 = 265150.$

Answer: a) 39973

b) 15999…99, with 224 consecutive digits 9

c) 265150

[Marking Scheme]

- **(a) 2 marks or 0, no partial score.**
- **(b) 2 marks or 0, no partial score.**
- **(c) 2 marks or 0, no partial score.**

6. Alam has 9 cards, where in each card, the numbers from 1, 2, 3, …, 9 are written.

1	2	3	4	5	6	7	8	9

At first, he takes one odd and one even numbered card. Afterwards, he again takes one odd and one even numbered card from the remaining cards. Now, he has four cards and then arranges them into a row. These four cards should have the following properties:

1) The product of each and every pair of odd and even cards selected is always a one-digit number.
2) The positive difference of the two odd cards and the two even cards must not be the same.

a) How many different 4-digit numbers can Alam form that satisfies the conditions above?
b) How many different 4-digit numbers can Alam form if the product in rule 1) is a two-digit number that is less than 20?
c) How many different 4-digit numbers can Alam form if the product in rule 1) is a multiple of 3 that is less than 40?

[Solution]

case a)
Regarding rule 1) the possibility for product of odd card and even card has 1-digit number is (1,2), (1,4), (1,6), (1,8), (2,3), so, the possible cards that Alam take is (1,4,2,3), (1,6,2,3), and (1,8,2,3). But regarding the rules 2), Alam card cannot be (1,4,2,3) since the difference of (1,3) and (2,4) is the same.
so, possibility of Alam card for case a) only (1,6,2,3), and (1,8,2,3) and with this Alam can make $2 \times 4! = 48$ arrangements

case b)
Regarding rule 1) the possibility for product of odd card and even card has 2-digit number less than 20 is (2,5), (2,7), (2,9), (3,4), (3,6), so, the possible cards that Alam take is (2,5,3,4), (2,5,3,6), (2,7,3,4), (2,7,3,6), (2,9,3,4), and (2,9,3,6). But regarding the rules 2), Alam card cannot be (2,5,3,4) and (2,7,3,6) since the difference of two odd cards and two even cards is the same.
so, possibility of Alam card for case b) only (2,5,3,6), (2,7,3,6), (2,9,3,4), and (2,9,3,6) and with this Alam can make $4 \times 4! = 96$ arrangements

case c)
Regarding rule 1) the possibility for product of odd card and even card has multiple of 3 less than 40 is (1,6), (2,3), (3,4), (3,6), (3,8),(4,9), (5,6) so, the possible cards that Alam take is (1,6,2,3), (1,6,3,4), (1,6,3,8), (1,6,4,9), (2,3,4,9), (2,3,5,6), (3,4,5,6), (3,6,4,9), (3,8,4,9), (3,8,5,6), (4,9,5,6). But regarding the rules 2), Alam card cannot

be (1,6,3,4), (1,6,3,8), (3,4,5,6), (3,8,5,6) and since the difference of two odd cards and two even cards is the same.
so, possibility of Alam card for case b) only (1,6,2,3), (1,6,4,9), (2,3,4,9), (2,3,5,6), (3,6,4,9), (3,8,4,9), (4,9,5,6) and with this Alam can make $7 \times 4! = 168$ arrangements

[Marking Scheme]

- **(a) 1 marks or 0, no partial score.**
- **(b) 2 marks or 0, no partial score.**
- **(c) 3 marks or 0, no partial score.**